

HAO LIU

Chicago, IL | +1 (312) 468-0476 | hliu232@uic.edu | liuhao030922.github.io

Education

University of Illinois Chicago

Ph.D. in Computer Science

GPA: 4.0/4.0; Focus: robotics, control, and multi-robot systems.

Chicago, IL

Aug 2025 – Present

Sichuan University

B.Eng. in Automation

GPA: 3.6/4.0 (86/100).

Chengdu, China

Sep 2021 – Jun 2025

Khalifa University

Exchange Student

Completed coursework and project collaboration in automation and intelligent systems.

Abu Dhabi, UAE

Aug 2024 – Dec 2024

Publications

Time-Aggregated Connectivity Maintenance for Multi-Robot Networks. *Robotics: Science and Systems (RSS)*, 2026.

Hao Liu, Yupeng Yang, Yanze Zhang, Yiwei Lyu, and Wenhao Luo.

Spatio-Temporal Reconnection for Multi-Robot Networks using Adaptive Prescribed-Time CBFs. *23rd IFAC World Congress (IFAC 2026)*, 2026.

Hao Liu, Yupeng Yang, Yanze Zhang, and Wenhao Luo.

Technical Skills

- **Programming:** Python, C++, Verilog, SQL
- **Tools:** ROS/ROS2, PyTorch, Git, Linux, MATLAB, Quartus, Arduino IDE, AutoCAD, Multisim

Awards

- Second-class Scholarship (top 20% students)
- Excellent Class Committee Member
- Outstanding Class Cadre

Selected Projects

Undergraduate Thesis: Vision-Array Reactor-Pool Intrusion Monitoring

Sep 2024 – Jun 2025

Computer Vision, Optical Flow, Edge Detection

- Designed an 8-camera vision-array layout for an irregular 11.77×10.12 m reactor-pool region, achieving full monitoring-area coverage under millimeter-level detection constraints.
- Built a detection pipeline combining attention-enhanced RAFT optical flow with Canny motion-region extraction; evaluated on 4 fastener types with ~ 100 drops each.
- Achieved 97%+ indirect ripple-based detection rates and ~ 0.15 s/frame on NVIDIA A100.

Legal Document Information Cleaner and Extractor

Sep 2022 – Jun 2023

Python, Regex, Data Processing

- Built a Python/regex pipeline to collect, normalize, and structure legal documents from public web sources.
- Designed extraction rules for key fields including charges and sentencing outcomes for downstream analytics.

Power Line Point Cloud Extraction and Fitting

Sep 2023 – Dec 2023

Python, DBSCAN, Least Squares

- Preprocessed noisy point clouds and applied DBSCAN to isolate power-line segments from mixed scenes.
- Used least-squares fitting to model line geometry for inspection and monitoring applications.

Assistive Exoskeleton Study for Muscular Dystrophy

Jan 2023 – Jun 2023

Gait Analysis, Wearable Systems

- Processed gait-related data and analyzed mobility-support requirements for an assistive exoskeleton system.
- Contributed to tension-adjustment strategy tuning to improve support across different movement phases.

Leadership

Student Union Executive Member

Sep 2021 – Jun 2022

- Planned student events including basketball tournaments and chess competitions.
- Coordinated with multiple departments to deliver cross-team campus activities.

Class Monitor

Sep 2021 – Jun 2022

- Organized monthly class activities, themed talks, and volunteer events.
- Managed class operations and communication between students and faculty.